PHS2000B: Exam 2

YOUR NAME HERE

Spring 2025

Welcome to the last exam of PHS2000B! As a reminder, you should complete and upload your exam to the Assignments tab on Canvas according to the schedule that you requested. You do not need to have Canvas open during the exam – just download the PDF file and upload your responses once you are done.

This exam has three sections, including a total of 36 points.

You may refer to your notes and any books you would like, but please do not discuss the exam with any other students from class or your tutor. By submitting this exam, you affirm that the work is yours alone and that you have not discussed the questions or responses with anyone other than the instructional team. By submitting this assignment, you also affirm that you did not use generative AI while completing this exam.

If you have clarifying questions for the instructional team, please email the whole team and we will get back to you as soon as possible. Please note that there is no class time devoted to working on this exam in person since the dates are so spread out.

There is a Google Doc where we will be posting any clarifications that arise from questions we receive during the exam. If we post a clarification, we will also post an Announcement referring you to the Google doc. Please note that any math or drawings you want to include can be typeset into the document or can be pasted in as an image. You are not required to typeset the math. Good luck!

Part 1: Regression Discontinuity Design/ Instrumetal Variable (10 points)

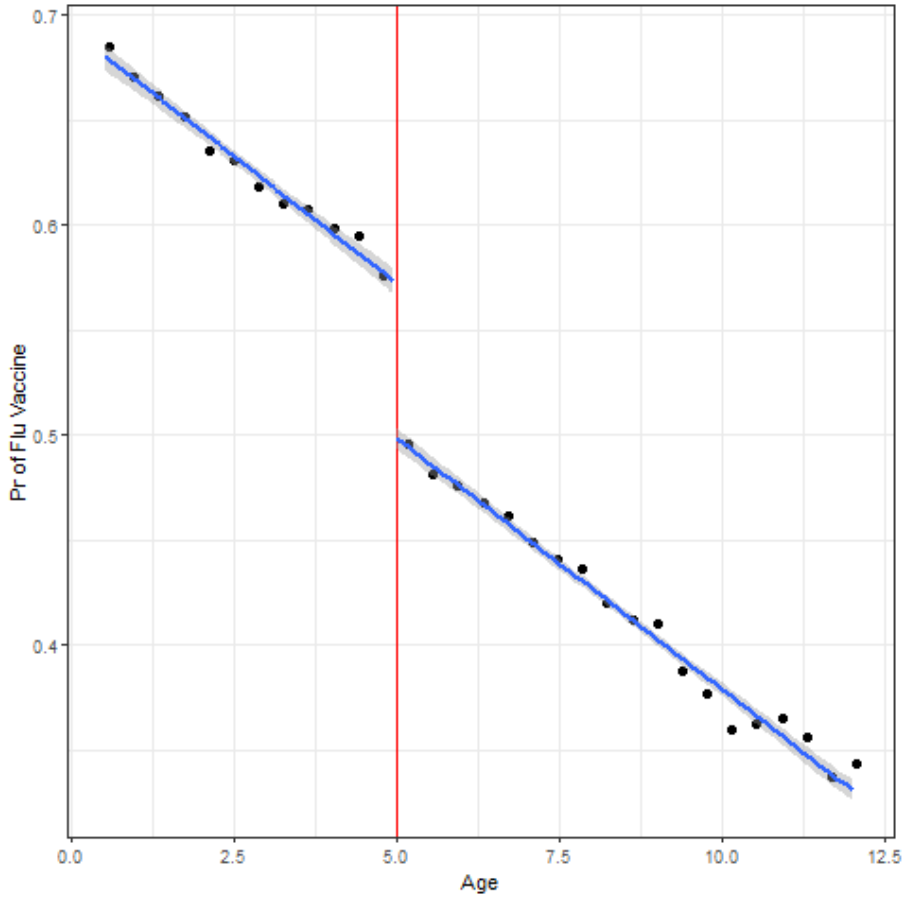
The country Jarvistan implemented a bold encouragement program regarding flu vaccination. Advertisements were aired on the radio and television everyday for two weeks warning of the risk of the flu infection for young children and encouraging parents to **vaccinate children under 5 years of age** to help avoid severe complications that can lead to emergency department (ED) visits. You are interested in evaluating the impact of vaccination on reducing ED visits. You have the below individual-level data from this past flu season for children under the age of 12 years with pre-existing comorbidities:

Variable Name	Variable Type	Explanation
age	Continuous	age in years
age_centered	Continuous	age in years **centered at the age 5 years**
under5	Dichotomous	Whether the child is aged less than 5 years (1= yes; 0= no)
flu_vacc	Dichotomous	Whether the child received the flu vaccine during the flu season in question (1= yes; 0= no)
ED_visit	Dichotomous	Whether the child had at least one ED visit (1 = yes; 0 = no)

Question 1

Identify the running variable, threshold, and treatment (1 point)

Question 2



Based on the graph, is this a fuzzy or sharp RDD? (1 point)

Question 3

Describe a defier in this context in words (1 point)

Question 4

```
> rdd_model1 <- feols(flu_vacc ~ age_centered + under5,
+                     data = subset(RDD_IV_sim, age_centered > -1 & age_centered <  1),
+                     se = "hetero")
>
> summary(rdd_model1)
OLS estimation, Dep. Var.: flu_vacc
Observations: 39,000
Standard-errors: Heteroskedasticity-robust
               Estimate Std. Error  t value   Pr(>|t|)
(Intercept)    0.501362   0.005526 90.73332  < 2.2e-16 ***
age_centered -0.029814   0.008939 -3.33514 8.5337e-04 ***
under5         0.070521   0.010078  6.99727 2.6520e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.496161   Adj. R2: 0.010195
```

```
> rdd_model2 <- feols(ED_visit ~ age_centered | flu_vacc ~ under5,
+                     data = subset(RDD_IV_sim, age_centered > -1 & age_centered <  1),
+                     se = "hetero")
>
> summary(rdd_model2)
TSLS estimation - Dep. Var.: ED_visit
                  Endo.    : flu_vacc
                  Instr.   : under5
Second stage: Dep. Var.: ED_visit
Observations: 39,000
Standard-errors: Heteroskedasticity-robust
              Estimate Std. Error  t value   Pr(>|t|)
(Intercept)   0.272292   0.053832  5.05816 4.2524e-07 ***
fit_flu_vacc -0.236510   0.100359 -2.35664 1.8446e-02 *
age_centered -0.014452   0.008975 -1.61022 1.0736e-01
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.347216   Adj. R2: 0.030776
F-test (1st stage), flu_vacc: stat = 49.2     , p = 2.342e-12, on 1 and 38,997 DoF.
                  Wu-Hausman: stat =  0.745643, p = 0.387864 , on 1 and 38,996 DoF.
> |
```

```
> rdd_model3 <- feols(ED_visit ~ age_centered + under5,
+                     data = subset(RDD_IV_sim, age_centered > -1 & age_centered <  1),
+                     se = "hetero")
>
> summary(rdd_model3)
OLS estimation, Dep. Var.: ED_visit
Observations: 39,000
Standard-errors: Heteroskedasticity-robust
              Estimate Std. Error  t value  Pr(>|t|)
(Intercept)   0.153715   0.003979 38.63141 < 2.2e-16 ***
age_centered -0.007401   0.006348 -1.16582  0.243696
under5       -0.016679   0.007214 -2.31201  0.020783 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.352657   Adj. R2: 1.634e-4
> |
```

From the model outputs above, select and interpret (1- 2 sentences) the estimates for the:

a) ITT at the threshold (1 point)

b) First-stage (1 point)

c) CACE (1 point)

Question 5

What assumptions are needed to interpret the above estimates causally? (2 points)

Question 6

Assume that providers also deliver a bundle of healthcare services like other required vaccines and health counselling when children come in for their flu vaccine. How does this affect what you are able to estimate in your causal inference and why? (2 points)

Part 2: Interrupted Time Series/ Difference-in-Differences (16 points)

Now consider a scenario where policy makers in Jarvistan instead roll out a huge effort to increase flu vaccination rates in April, 2022. The campaign lasted for six months. One policymaker took PHS2000B (what a small world!) and suggests using an ITS analysis to evaluate the effect of the program on the number of ED visits for children under 5 years. You leverage monthly aggregate data of all ED visits for children under 5 years, with data ranging from January 2021 to December 2023.

Variable Name	Variable Type	Explanation
month_centered	Continuous	Measure of time in months, centered at April 2022
inter_first_6mo	Dichotomous	Whether the month is during the first 6 months after the campaign (1= yes; 0= no)
monthly_ED_visits	Continuous	Number of ED visits for children under 5

Question 1

What assumption(s) are needed to interpret ITS estimate causally? Explain said assumption(s) in the context of this research design. Choose one assumption and hypothesize a possible violation (3 points)

Question 2

```
> its_model <- feols(monthly_ED_visits ~ month_centered + inter_first_6mo, data=its_data, se = "hetero")
> summary(its_model)
OLS estimation, Dep. Var.: monthly_ED_visits
Observations: 36
Standard-errors: Heteroskedasticity-robust
                 Estimate Std. Error  t value   Pr(>|t|)
(Intercept)     1274.3712   25.67830 49.62833  < 2.2e-16 ***
month_centered    29.0248    2.35579 12.32062 6.8302e-14 ***
inter_first_6mo -140.4333   34.92649 -4.02083 3.1713e-04 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 122.7   Adj. R2: 0.853191
```

Which of the below hypothesized effect types is estimated in the above model? Explain why. (2 point)

1) Level change at time of intervention that persists throughout post-period
2) Slope change at the time of the intervention that persists throughout post-period
3) Level and slope change at time of intervention that persists throughout post-period
4) Lagged slope change that persists throughout post-period
5) Temporary level change that does NOT persist throughout post-period
6) Temporary slope change that does NOT persist throughout post-period

Question 3

Interpret the parameters estimated for the *intercept*, *month_centered*, & *inter_first_6mo* (1- 2 sentences each) (3 points)

Question 4

Due to the high visibility of this vaccine campaign and the scrutiny on its success, healthcare providers improve their record-keeping practices for healthcare encounters for children, which includes vaccinations and ED visits. How does this affect your confidence in the assumptions needed for ITS? Explain why. (2 points)

Questions 5-8

You learn that there is another country, McConnelland, that did not have the same kind of vaccine campaign. Fortunately, you also have individual-level panel data for children under 5 from Jarvistan and McConnelland. As such, you are able to conduct a Difference-in-Difference analysis

Variable Name	Variable Type	Explanation
Post	Dichotomous	Is an observation from April 2022 or after (1= yes; 0= no)
Treat	Dichotomous	Is an observation from a (1= yes; 0= no)
ED_visit	Dichotomous	Whether the child had at least one ED visit (1 = yes; 0 = no)

Question 5

```
> did_model <- feols(ED_visit ~ treat*post, data=did_data, se = "hetero")
> summary(did_model)
OLS estimation, Dep. Var.: ED_visit
Observations: 720,000
Standard-errors: Heteroskedasticity-robust
             Estimate Std. Error  t value  Pr(>|t|)
(Intercept)  0.054860   0.000588 93.30915 < 2.2e-16 ***
treat        0.049007   0.000983 49.85656 < 2.2e-16 ***
post         0.056235   0.000903 62.25626 < 2.2e-16 ***
treat:post  -0.007402   0.001433 -5.16648 2.386e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.311019   Adj. R2: 0.011981
```

Interpret the primary causal estimate of interest (1- 2 sentences) (1 point)

Question 6

For each of the below statements, indicate whether the statement violates a necessary identification assumption for DiD and why or why not this assumption is violated (3 points)

a) Jarvistan's under-5 hospitalization rate has been higher than McConnelland in the five years preceding the rollout of the vaccination campaign.

b) Flu season is a different time of year between the two countries

c) Lower flu transmission in Jarvistan leads to less flu cases in McConnelland as they share a border and frequent migration back and forth.

Question 7

What visual evidence could you prepare that might assure a reader that McConnelland would be likely to be a good comparison for the outcomes of Jarvistan? (1 point)

Question 8

Suppose visual evidence suggests that McConnelland may NOT be an appropriate comparison for the outcomes in Jarvistan. Imagine that you have data from 10 other similar countries – describe what you might do to construct a proper counterfactual. (1 point)

Part 3: Selection Bias and Missing Data (10 points)

Throughout this part, please ignore issues related to interference, where one person's treatment can affect the potential outcomes of another person.

Questions 1-2

Your team is interested in exploring the causal effect of flu vaccination on childhood mortality. Your colleague proposes the following large-scale study using electronic health record data: You will (1) include infants and children that show up to the pediatric ER for respiratory-related issues, (2) determine whether those children have received a flu vaccine in the past via their medical record, and (3) follow their mortality outcomes using the national death registry. To determine whether this is a good design, you consider the following questions.

Question 1

Draw a DAG that depicts your study. Your DAG should include **only** the following 4 nodes: flu vaccination, an indicator of whether the ER visit was respiratory in nature, mortality, and a node depicting whether the child was born preterm (unmeasured but hypothesized to influence ER visits and mortality). For the purposes of this question, assume that flu vaccination is as good as random. **(1 point)**

Question 2

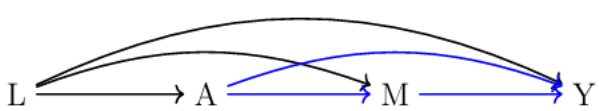
Given the study design and your corresponding DAG, determine which of the following causal effects are identifiable using only data from infants and children included in your study. Explain each answer in 1-2 sentences.

a) The average causal effect of flu vaccination on childhood mortality in your study sample. **(1 point)**

b) The average causal effect of flu vaccination on childhood mortality in the full population of pediatric ER visitors. **(1 point)**

Questions 3-5

As an alternative to the previous approach, you propose a different study. Using a large cohort of children, you will leverage existing longitudinal data on respiratory illness and flu vaccination during childhood. Participants consented to providing their unique personal identification number, which enables linkage to future medical and vital records. You begin with the following DAG, where L = wheezing age (0-6 mo), A = flu vaccine (6-12 mo), M = flu and related complications, and Y = mortality. You are interested in the total effect of A on Y (i.e., the blue arrows).



However, you soon learn that only 60% of your participants consented to record linkage–that is, you are not able to ascertain mortality outcomes for 40% of your study sample. Exploratory analyses suggest that parents whose children wheezed before the age of 6 months were less likely to consent to record linkage. Parents who vaccinated their children for the flu between 6 and 12 months were more likely to consent to record linkage.

Question 3

Using this additional information, update your DAG. **(1 point)**

Question 4

Assuming that early-life wheezing modifies the relationship between flu vaccine and mortality, which of the following causal estimands are identifiable using a complete case analysis? While you may defend your answer either using words or math, you must clearly specify key independence statement(s) that support your claim (i.e., of the form $X_1 \perp\!\!\!\perp X_2$). Recall that L = wheezing age (0-6 mo), A = flu vaccine (6-12 mo), M = flu and related complications, and Y = mortality.

a) Conditional average treatment effect in the selected population (i.e., the complete cases): $E[Y^{a=1}|L = l, R_Y = 1] - E[Y^{a=0}|L = l, R_Y = 1] = Pr[Y^{a=1} = 1|L = l, R_Y = 1] - Pr[Y^{a=0} = 1|L = l, R_Y = 1]$ **(1 point)**

b) Conditional average treatment effect in the target population (i.e., the full sample): $E[Y^{a=1}|L = l] - E[Y^{a=0}|L = l] = Pr[Y^{a=1} = 1|L = l] - Pr[Y^{a=0} = 1|L = l]$ **(1 point)**

c) Marginal average treatment effect in the complete cases: $E[Y^{a=1}|R_Y = 1] - E[Y^{a=0}|R_Y = 1] = Pr[Y^{a=1} = 1|R_Y = 1] - Pr[Y^{a=0} = 1|R_Y = 1]$ **(1 point)**

d) Marginal average treatment effect in the target population (i.e., the full sample): $E[Y^{a=1}] - E[Y^{a=0}] = Pr[Y^{a=1} = 1] - Pr[Y^{a=0} = 1]$ **(1 point)**

Question 5

Would multiple imputation help you obtain an unbiased estimate of the marginal average treatment effect in the target population (i.e., the full sample)? While you may defend your answer in words, you must clearly specify the missingness structure and key independence statement(s) that support your claim (i.e., of the form $X_1 \perp\!\!\!\perp X_2$). **(2 points)**